Analog LED dimming — simulation check of the rev 2 design

Trenholm lab · behaviour-chamber lighting · 14 Oct 2026 (rev d) · ngspice 42 + Monte Carlo, run against the circuit in *analog_dimming_rev2.pdf*

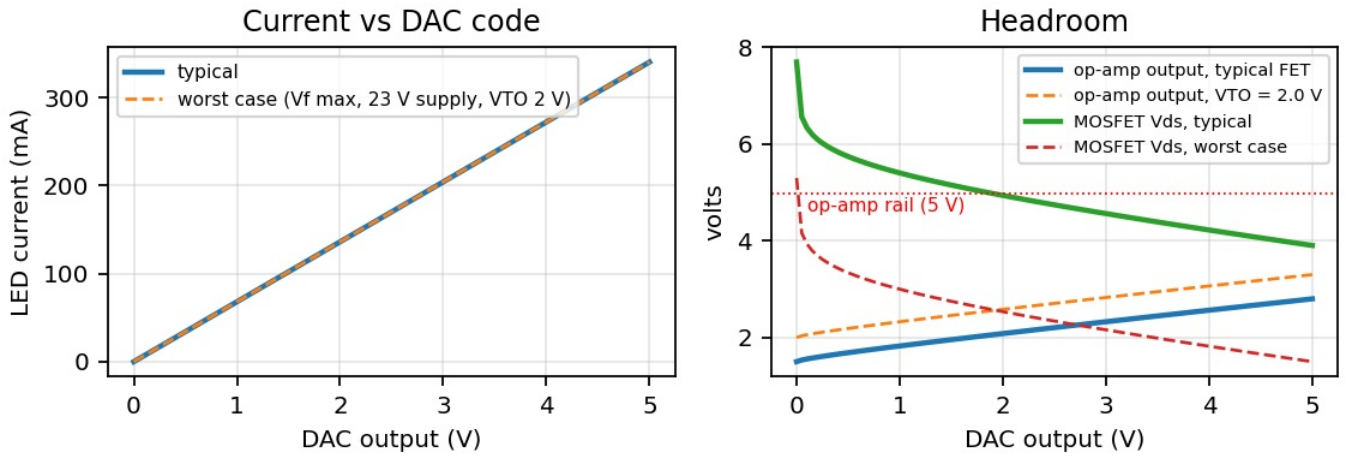
Verdict (rev b, after the bench test of 13 Oct). The analog current-sink circuit (Adafruit GP8403 DAC → 3.9 k/1 k divider → OP295 → 1 kΩ → IRLZ44N → sense resistor) passes every check: the current is set by the DAC code and by nothing else, it stays set with the worst-case MOSFET, the worst-case LED voltage and a low supply, the loop has ~90° of phase margin (it cannot oscillate), it steps between levels without overshoot, and the MOSFET never runs hot. **The one open question — how the DuoStrip's on-board regulator behaves when starved of voltage — has now been answered on the bench: a spare segment dims smoothly and steadily from 24 V down to ~13.5 V with no cut-out, which is the "linear / pass-through" case (fig. 4, left).** That removes the only reason to replace the LEDs. **Order the DuoStrip version of the circuit (section 8); keep the existing strips; no Bridgelux modules needed.**

What the bench test settled. Before 13 Oct the DuoStrip variant could not be guaranteed because the strip's driver chip is undocumented: a chip with an under-voltage lockout would turn the 10-lux setting into a ~14 kHz hiccup (fig. 4, right). On a variable supply a spare segment instead dimmed continuously down to ~13.5 V — about 2 V below where any plausible lockout would have cut it off — with steady light at every fixed voltage. (The slight flicker while turning the knob is the supply's output jitter amplified by the LEDs' steep current–voltage curve; the sink circuit holds current, not voltage, and filters the set-point at 20 Hz, so that mechanism does not exist in the real circuit.) A lockout chip cannot produce a smooth fade through 16 V, so this by-eye result is conclusive for the question that mattered.

1 · What was simulated

The whole chain from DAC output to LED string was built as a SPICE netlist: an OP295-like op-amp macromodel (100 dB gain, 75 kHz gain-bandwidth, output clamped 0.02–4.98 V on 5 V, 50 Ω output), the 3.9 k/1 k divider with its 10 µF filter, the 1 kΩ gate resistor, the 1 kΩ + 10 nF compensation, an IRLZ44N (level-1 MOSFET with explicit 1.5 nF gate capacitance; also run with 3 nF and a 2.0 V threshold, the datasheet maximum), the 3.0 Ω sense resistor, and a Bridgelux EB 280 mm module modelled as 6 series LEDs × 9 parallel strings, fitted to 19.1 V at 350 mA. A 300 µV op-amp input offset (OP295 maximum) was included in every run. Worst case adds +1.4 V to the module (Bridgelux's maximum Vf bin) and lowers the supply to 23.0 V.

2 · DC transfer and headroom

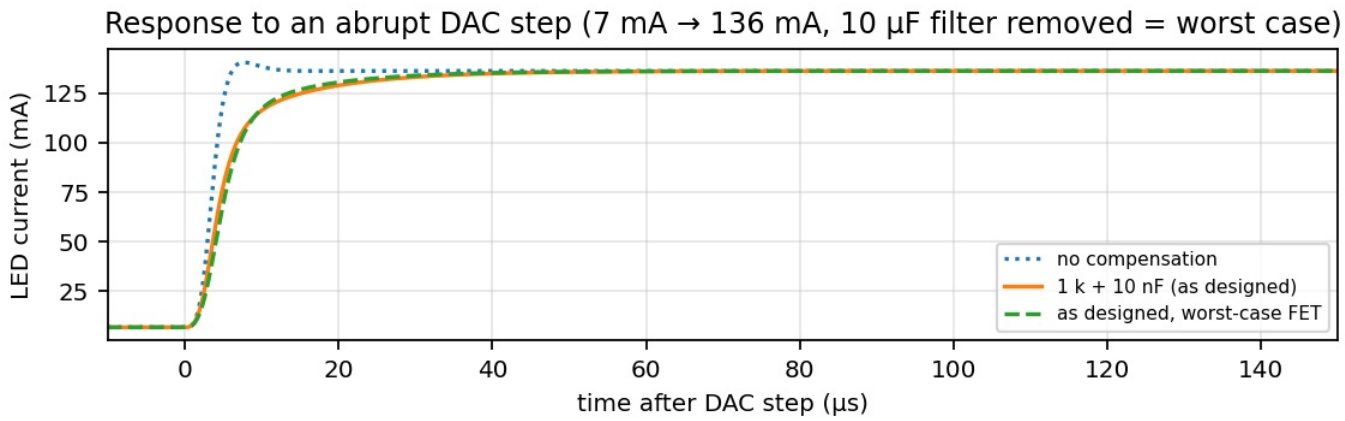


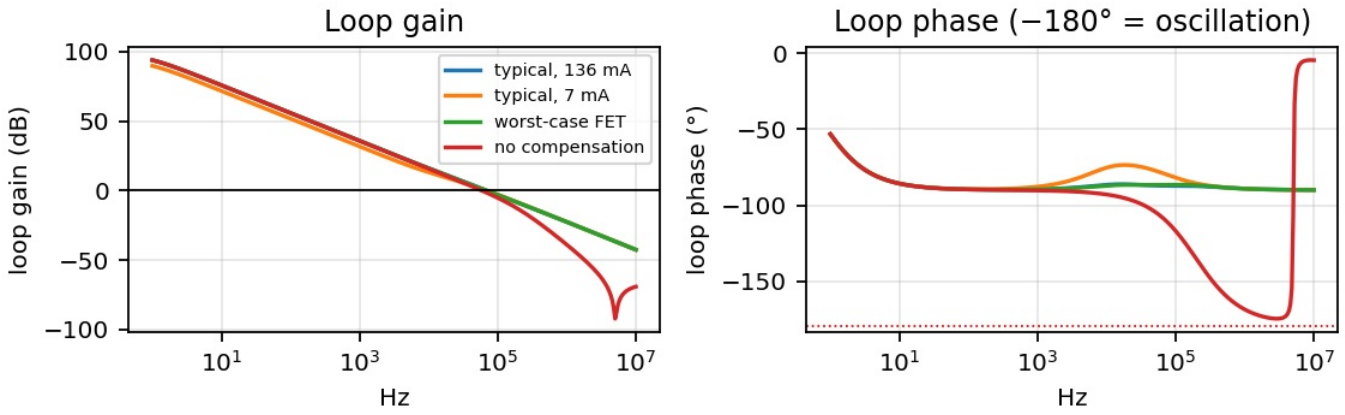
DAC volts	LED current (per module)	op-amp output needed (typ / VTO = 2 V)	MOSFET Vds (typ / worst)	MOSFET power
0.00	0.09 mA (offset only; → 0 with code 0, see note)	1.5 V / 2.0 V	7.7 V / 5.3 V	0
0.05 (≈10 lux)	3.5 mA	1.5 / 2.0 V	6.6 / 4.2 V	0.02 W
0.20 (100 lux)	13.7 mA	1.6 / 2.1 V	6.1 / 3.7 V	0.08 W
2.00 (≈1000 lux)	136 mA	2.1 / 2.6 V	4.9 / 2.5 V	0.7 W
5.00 (full scale)	340 mA	2.8 / 3.3 V	3.9 / 1.5 V	1.3 W

Current is a straight line in DAC code (0.083 mA per count) and the **typical and worst-case curves lie on top of each other** — the module voltage, supply voltage and MOSFET threshold have no effect on the current, which is the whole point of the design. The op-amp needs at most 3.3 V of its 4.98 V swing (the LT1013 in rev 1 would have run out here; the OP295 does not). The MOSFET keeps ≥1.5 V of drain–source voltage even in the worst corner, so it stays in regulation. Peak dissipation 1.3 W at full scale, 0.7 W at 1000 lux — a clip-on TO-220 heatsink (≤25 °C/W) holds the tab under 60 °C.

Note on "off": with a +300 µV offset the loop would try to sink 0.1 mA at DAC code 0; the OP295's output cannot go below ~20 mV so in practice the MOSFET is fully off (gate ≈ 0 V ≪ threshold). Bench check A in the rev 2 sheet (multimeter across the sense resistor reads 0 mV at code 0) confirms this on the real parts.

3 · Stability: step response and loop gain





Case	Overshoot	Settles to 1 %	Crossover	Phase margin
As designed (1 kΩ + 10 nF), typical FET, 136 mA	0 %	40 µs	69 kHz	**93°**
As designed, worst-case FET (3 nF gate, VTO 2 V)	0 %	40 µs	~70 kHz	~92°
As designed, 7 mA (near the 10-lux point)	0 %	—	~45 kHz	101°
No compensation (for comparison)	3.2 %	60 µs, 5 ring cycles	~100 kHz	73°

Anything above ~60° is considered unconditionally stable; the design sits at 90–100° over the whole current range, including with the worst MOSFET in the bin. The step test deliberately removes the 10 µF filter so that the loop sees an instantaneous DAC edge — the real circuit sees a 20 Hz-filtered ramp and the loop never even notices a level change. There is no mechanism for flicker: no switching, no oscillation, a 100-dB DC loop gain, and a 20 Hz low-pass between the DAC and the control input.

4 · DAC ripple

The GP8403 datasheet does not state its internal PWM frequency, but it specifies a 100–200 µs output response time, which puts its internal filter corner in the kHz range and its PWM well above 10 kHz. The 3.9 k/1 k + 10 µF network is a 20 Hz low-pass (Thevenin 796 Ω × 10 µF), so any residual ripple at the DAC pin is attenuated by at least 500× before the op-amp, and the op-amp's 75 kHz bandwidth adds another roll-off. Even a pessimistic 20 mV of DAC ripple becomes <0.01 mA of LED ripple — below the 12-bit step and invisible to any camera or mouse.

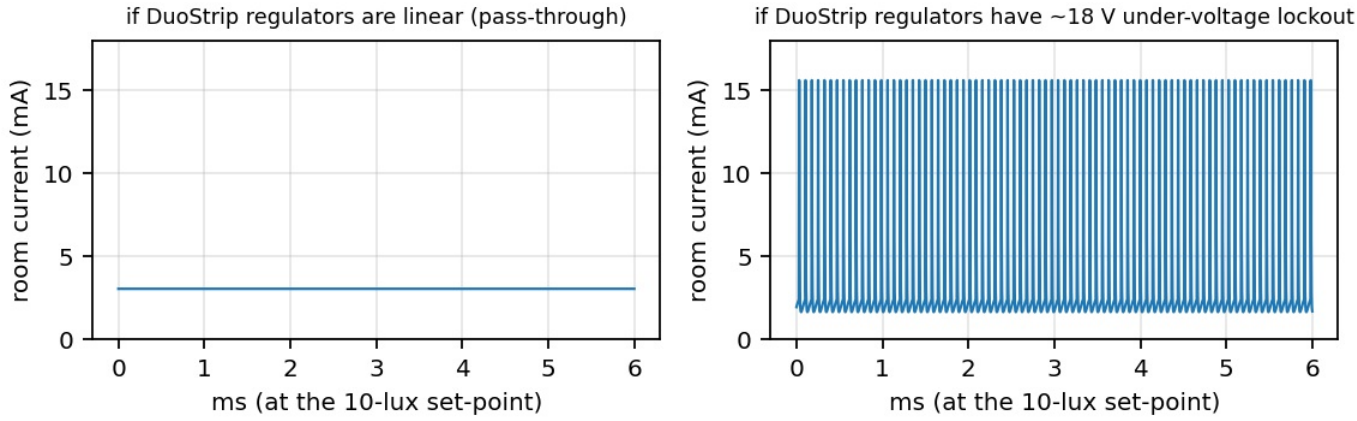
5 · Current sharing between parallel LED strings (Monte Carlo)

A current sink controls the *total* current; how it divides between the parallel strings inside a module depends on string-to-string forward-voltage mismatch. This matters at 10 lux, where each string carries only ~0.15 mA and the exponential diode law makes small voltage differences into large current differences. 400 random modules were drawn per case (Nichia/Bridgelux-class binning: σ = 30 mV per LED typical, 60 mV pessimistic).

Case	Total current	brightest / dimmest string (median)	95th percentile	brightest / mean
Bridgelux EB, 9 strings × 6 LEDs, one board, typical binning	1.35 mA (10 lux)	2.0×	2.8×	1.36×
	13.5 mA (100 lux)	1.9×	2.8×	1.34×
	135 mA (1000 lux)	1.7×	2.3×	1.28×
Bridgelux EB, pessimistic binning	1.35 mA	4.0×	8.2×	1.8×
DuoStrips, 16 strings over 2 reels (if regulators are linear)	3.2 mA (10 lux)	2.6×	4.6×	1.56×
	320 mA (1000 lux)	1.6×	2.2×	1.25×
DuoStrips, pessimistic (reels 150 mV apart)	3.2 mA	6.4×	17×	2.2×

What this means for the experiment: the *total* light on the floor is set by the total current and is unaffected — the sink holds it to 12-bit precision regardless of how the strings share. The sharing affects only the *spatial* pattern along the 280 mm bar, by a fixed amount that does not drift (it is baked into the LED bins). For a 280 mm bar 40 cm above a 30 cm floor the floor illuminance integrates over the whole bar, so a 2× brightest-to-dimmest string spread translates to roughly ±10 % across the floor at 10 lux — the same order as the geometric fall-off from centre to corner you already have. Nothing to act on; it is worth knowing that at 10 lux the bar will look slightly "patchy" to a camera even though the lux meter reads exactly what you set. (The PWM scheme hid this because the LEDs always ran at full current when on.) The DuoStrip numbers are worse because its 16 strings come from two reels, which is one more reason to prefer the single-board Bridgelux modules.

6 · DuoStrip variant (Option 2): linear vs under-voltage-lockout regulator



Same sink, 2.0 Ω sense, driving one room of 8 DuoStrip segments (16 strings) modelled with their on-board regulator in two ways. **Left:** regulator as a linear current limiter that simply passes whatever current the sink allows — the room sits at a rock-steady 3.05 mA at the 10-lux set-point, 30.6 mA at 100 lux, 306 mA at 1000 lux, zero ripple; Vds on the MOSFET 6.8 / 5.9 / 3.8 V, dissipation ≤1.2 W. **Right:** the same regulator with an 18 V under-voltage lockout (0.5 V hysteresis). At 10 lux the strip's terminal voltage must fall to ~17 V for the sink to reach its set-point, the regulator switches off, the sink opens the MOSFET fully, the strip voltage recovers, the regulator switches on, and so on at ~14 kHz: the room current flickers between 1.7 and 15.6 mA (average 3.3 mA — a lux meter would read "correct", the camera and the mouse would not agree). The 100 and 1000 lux points clear the lockout in this run by only ~0.5 V; with a 19 V lockout the 100-lux point fails too. **Bench result (13 Oct):** a spare segment on a variable supply dimmed smoothly and steadily down to ~13.5 V with no cut-out — the left panel is the real DuoStrip. The simulated DuoStrip room therefore behaves exactly like the Bridgelux case: 3.05 / 30.6 / 306 mA at the three set-points with zero ripple, MOSFET Vds 6.8 / 5.9 / 3.8 V, and ≤1.2 W in the MOSFET at 1000 lux (hence the bolt-on ≤10 °C/W heatsink in the DuoStrip parts list rather than a clip-on).

7 · Residual risks and how they are covered

Risk	Status
GP8403 library on a Pi 5	Adafruit's `adafruit-circuitpython-gp8403` runs through Blinka, which supports the Pi 5's I²C (it does not touch RPi.GPIO). The DFRobot library was rejected for exactly this reason.
Op-amp supply	OP295 V+ from the Pi's 5 V pin (pin 2 or 4; one female-to-male jumper), V− to the local ground rail. Draws <1 mA. 0.1 µF across the pins.
Soldering /	None: the 6516 DAC ships with STEMMA QT and screw terminals; op-amp, resistors and capacitors go on a mini solderless

PCB	breadboard; sense resistors and MOSFET leads in WAGOs; Bridgelux modules have poke-in connectors.
Resolution at 10 lux	~16 DAC counts → each count is ~6 % of 10 lux (≈0.6 lux). You said ±2 lux is acceptable; this is three times finer than that, and unlike the PWM cliff it is monotonic and repeatable.
Long-term drift	Current is set by two 1 % resistors, a 3.0 Ω 1 % sense resistor and the DAC reference. Metal-film tempco ≈ 50 ppm/°C → <0.2 % over a 20 °C room swing. OP295 offset drift 1–3 µV/°C → 0.1 %/°C at 10 lux, 0.01 %/°C at 100 lux. LED aging is the only remaining drift and is slow (the Bridgelux module at 135 mA runs at ~40 % of rating, ~5 % lumen loss expected over several years).
Off state	Code 0 gives gate ≈ 0 V; MOSFET fully off; verify with the multimeter (check A). If you need "really off" for an IR-only phase you can also cut the DAC to 0 and the LEDs will be dark to better than 0.01 mA.
Bridgelux part availability	BXEB-L0280Z-50G1000-C-C3 (CRI 90) preferred; BXEB-L0280Z-50E1000-C-C3 (CRI 80) is the drop-in fallback with the same voltage and current.

8 · Order list with prices — DuoStrip version (keep the existing LEDs)

This is the parts list from *duostrip_analog.pdf*, priced. CAD per unit, 13 Oct 2026. The two Adafruit prices are live-checked (US$7.50 and US$0.95 on adafruit.com, ×~1.37); Digi-Key Canada lines are typical list prices (their pages don't expose prices to my reader) and may land ±30 % — **confirm in the cart**. One Digi-Key Canada order covers everything (they stock the Adafruit parts too). No LEDs, no MOSFETs (you reuse the two IRLZ44N), no PCB, no soldering.

Qty	Part	Source	Each (CAD)	Line (CAD)
1	Adafruit 6516 — GP8403 2-channel I²C DAC, 0–5 V range (screw terminals, STEMMA QT)	adafruit.com/product/6516 or digikey.ca	~10.50	10.50
1	Adafruit 4397 — STEMMA QT to female-socket cable (plugs onto Pi pins 1/3/5/6)	adafruit.com/product/4397 or digikey.ca	~1.50	1.50
2	OP295GPZ dual rail-to-rail op-amp, DIP-8 (one chip runs both rooms; second is a spare)	digikey.ca 820348	~9.50	19.00
3	2.0 Ω 2 W metal-film / metal-oxide resistor, axial (sense; one per room + spare)	digikey.ca	~0.70	2.10
2	TO-220 bolt-on heatsink ≤10 °C/W, with M3 screw and nut (for the existing MOSFETs)	digikey.ca	~2.50	5.00
2 / 4	3.9 kΩ ¼ W 1 % · 1.0 kΩ ¼ W 1 % (divider bottom + feedback per room; existing gate resistors reused — buy a 10-pack of each)	digikey.ca / any	~0.15	2.00
2 / 2 / 1	10 µF X7R leaded ceramic · 10 nF C0G leaded · 0.1 µF ceramic	digikey.ca / any	0.20–0.80	2.50
2	WAGO 221-413 (3-way) — one per room (source wire + 2.0 Ω + sense wire)	have / digikey.ca	~0.90	1.80
1	Mini solderless breadboard + jumper wires (incl. 1 female-to-male for the Pi 5 V pin)	you have these	—	0
			Total, both rooms (before shipping/tax)	**~C$44**

Shipping from Digi-Key Canada is typically ~C$8. Software: `chamber_lights_duostrip.py` — run `sweep()` once per room with the lux meter to fill in the three calibration currents (starting estimates 320 / 32 / 3.2 mA per room).

9 · Wiring the new components — minimum-change version

Based on your schematic (Pi GPIO → series resistor R19–R22 → IRLZ44N gate; all four sources on the common ground net; Q2 and Q3 driving the two white-LED rooms; Q1 and Q4 the IR banks, which stay exactly as they are). There is no separate gate-driver board to remove, and the existing gate resistors are reused as the circuit's gate resistors, so the total change on the existing wiring is **one wire moved and one wire lifted per room**. Everything else is new parts on the breadboard, which touches the existing build at only three points: the ground net, a 5 V pin and the I²C pins of the Pi.

> **Same topology as your schematic: low-side.** In your build the MOSFET sits between the LED return and ground (drain on the LED cathode bus, source on the ground net) and the +24 V rail feeds the LED anodes directly. The analog circuit keeps exactly that arrangement. The MOSFET never touches +24 V; the only change on the power path is a 2.0 Ω resistor between the source and ground, and the only change on the control path is what drives the gate resistor (the op-amp instead of the GPIO). The drain, the LED return bus and the +24 V feed are not touched.

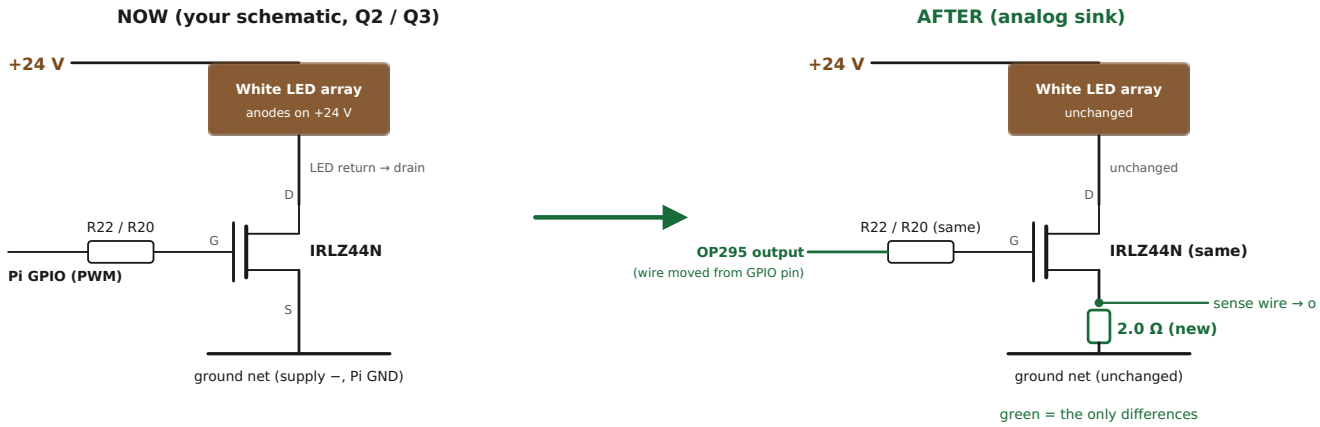


Figure 5a. Before and after for one room. Low-side in both: the MOSFET gates the LED return to ground; +24 V goes straight to the LED anodes and is never switched. The sink adds the 2.0 Ω between source and ground and replaces the GPIO with the op-amp as the thing driving the existing gate resistor.

Per room (Q2 for room 1, Q3 for room 2):

1. **Gate — move one wire.** Follow the gate resistor's Pi-side wire back to the header and unplug it from its GPIO pin. Plug that same wire into the breadboard row of the op-amp output: OP295 pin 1 for room 1, pin 7 for room 2. The resistor (R22 / R20) stays in place and is now the gate resistor of the current sink. Any value up to 2.2 kΩ is fine as-is (simulated: no overshoot, ~93° phase margin from 47 Ω to 2.2 kΩ); if the colour bands read 4.7 kΩ or more, replace it with 1 kΩ. The freed GPIO pin is no longer used by the lights.
2. **Source — lift one wire and add the sense resistor.** Pull the MOSFET's source wire out of the ground net (the WAGO or bus where it lands now). Push the 2.0 Ω resistor's bottom lead into that vacated slot — a WAGO 221 clamps a solid resistor lead fine. Put the source wire, the 2.0 Ω's top lead and one new wire (the "sense" wire) together in one new WAGO 221-413. The sense wire goes to the breadboard, to the feedback 1 kΩ. Taking it from the WAGO rather than the breadboard keeps breadboard contact resistance out of the measurement.
3. **Heatsink.** Bolt the ≤10 °C/W heatsink to the MOSFET tab with the M3 screw and a dab of thermal compound. The tab is the drain, several volts above ground: it must not touch the other MOSFET's heatsink, the chassis or any ground wiring.

Once, shared by both rooms:

4. **Breadboard.** Seat the OP295 across the centre channel, notch toward you (pin 1 bottom-left, 1–4 down the left, 5–8 up the right). Per room: 3.9 kΩ + 1 kΩ divider, 10 µF, 1 kΩ feedback, 10 nF. Once: 0.1 µF between pin 8 and the ground rail. Wire it row by row from the connection list below; each row is one node.
5. **Ground rail → ground net.** One wire from the breadboard ground rail to the same ground net the 2.0 Ω resistors now land in — directly, not via the Pi. (The DAC's ground also reaches the Pi through the STEMMA cable; that is fine, but the op-amp's reference has to be the resistors' ground end, which this wire guarantees.)
6. **Op-amp supply.** One female-to-male jumper from a free Pi 5 V pin (pin 2 or 4) to OP295 pin 8; a jumper from the ground rail to pin 4.
7. **DAC.** STEMMA QT cable from the GP8403 to Pi pins 1 (3.3 V, red; pin 17 is the alternative 3.3 V if pin 1 is taken), 3 (SDA, blue), 5 (SCL, yellow), 6 (GND, black) — your schematic shows pins 3 and 5 unused. Screw terminals: VOUT0 → room-1 3.9 kΩ, VOUT1 → room-2 3.9 kΩ, GND → breadboard ground rail.
8. **One-room trial.** Do room 1 first and leave room 2 on PWM; jumper the unused OP295 half (pin 5 → ground rail, pin 6 → pin 7) until room 2 is converted. Boot the Pi, `i2cdetect -y 1` shows the DAC at 0x58–0x5F, then power the 24 V and run the six checks at the end of this section.

New wires in total: per room, one moved (gate) and two new (sense wire, plus the source wire now terminating in the new WAGO); shared, five short jumpers (ground rail, 5 V, two DAC outputs, DAC GND) and the STEMMA cable. The 24 V feed, the strips, the drain wiring, the IR channels, the fuse and the supply are untouched.

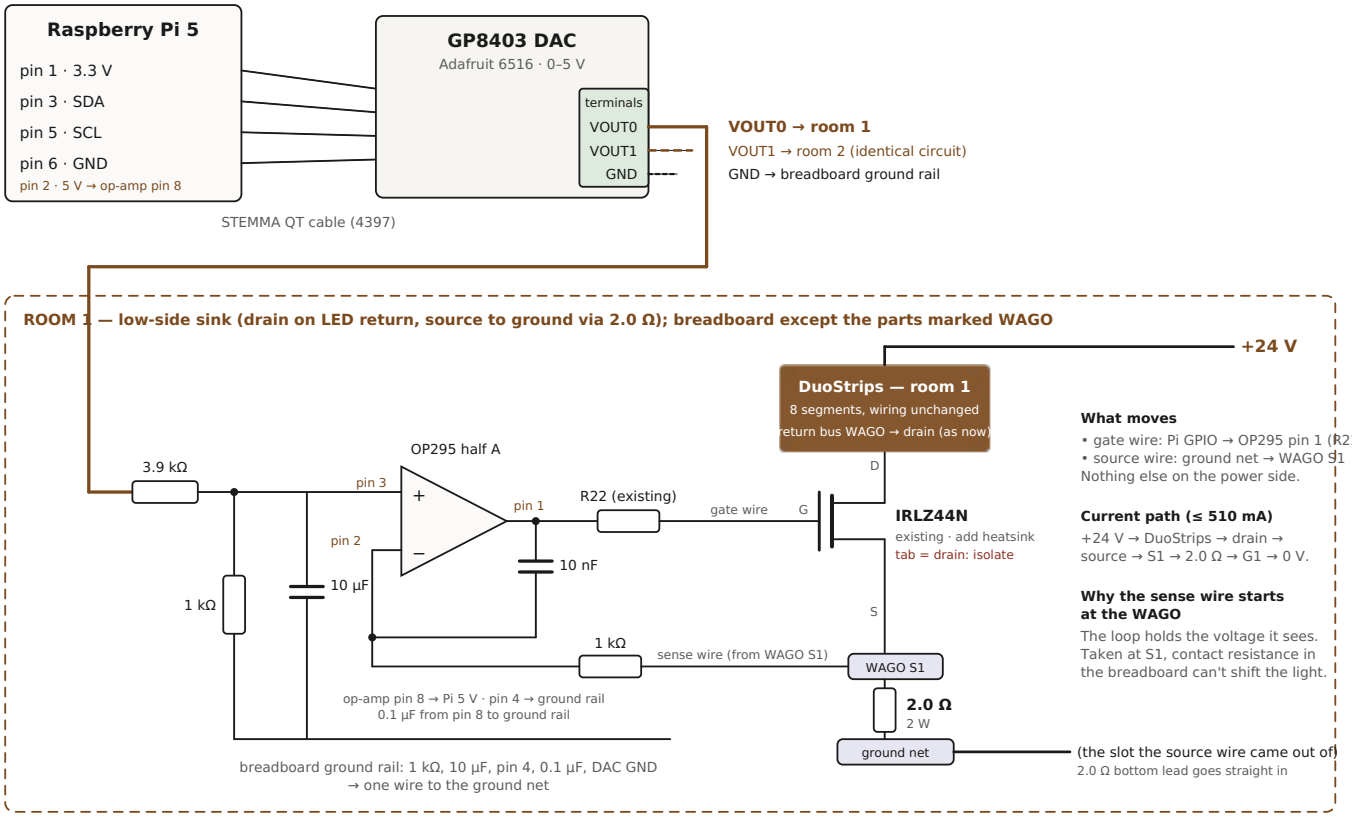


Figure 5b. Room 1 wiring in full (low-side, as above). Room 2 is identical using VOUT1 and the other half of the same OP295 (pin 5 = +, 6 = −, 7 = output), with its own divider, capacitors, sense resistor and WAGOs.

Connection list — check every wire against it. OP295 pin 1 is next to the notch/dot.

Node	Room 1 — connect together	Room 2 — connect together
I²C	STEMMA QT cable from DAC → Pi pin 1 (3.3 V), pin 3 (SDA), pin 5 (SCL), pin 6 (GND). Red = 3.3 V, black = GND, blue = SDA, yellow = SCL.	
+5 V	Pi pin 2 · OP295 pin 8 · one leg of 0.1 µF	
GND rail	OP295 pin 4 · other leg of 0.1 µF · DAC GND terminal · both 1 kΩ divider-bottom resistors · both 10 µF · *one wire to the ground net*	
DAC out	DAC VOUT0 · 3.9 kΩ (end a)	DAC VOUT1 · 3.9 kΩ (end a)
SET	3.9 kΩ (end b) · 1 kΩ divider bottom (other end on GND rail) · 10 µF (other leg on GND rail) · OP295 pin 3	same, with OP295 pin 5
FB	OP295 pin 2 · 1 kΩ feedback (end a) · 10 nF (leg a)	OP295 pin 6 · 1 kΩ feedback (a) · 10 nF (a)
OUT	OP295 pin 1 · 10 nF (leg b) · the Q2 gate wire that used to go to the Pi GPIO	OP295 pin 7 · 10 nF (b) · the Q3 gate wire that used to go to the Pi GPIO
GATE	Unchanged — existing R22 / R20 to the MOSFET gate (≤2.2 kΩ; swap for 1 kΩ only if ≥4.7 kΩ)	
WAGO S1 / S2 (new)	Q2 source wire · 2.0 Ω top lead · sense wire to 1 kΩ feedback (end b)	Q3 source wire · 2.0 Ω top · sense wire to 1 kΩ feedback (b)
Ground net	Existing common ground (Pi GND, supply −, IR MOSFET sources) · 2.0 Ω bottom leads, in the slots the source wires vacated · one wire from the breadboard ground rail	
Drain, +24 V, IR	Unchanged — drains stay on their rooms' return buses; Q1/Q4 and the IR banks are not touched.	

Trial on room 1 — pass all six before converting room 2.

1. **Off is off.** Run the script's `off()`. LEDs dark; multimeter across the 2.0 Ω reads ~0 mV. Power-cycle everything; still dark before any script runs.
2. **Current tracks the command.** `set_current_mA(1, 300)` → 0.60 V across the 2.0 Ω (±3 %). `set_current_mA(1, 30)` → 60 mV.
3. **Corners match at the dim end** — the key test for the DuoStrips. Lux at all four floor corners and centre, at the 1000 lux and 10 lux settings. The corner-to-centre ratios should be similar at both. If one corner is more than ~1.5× off at 10 lux but fine at 1000, the strips aren't sharing current evenly when starved — stop and tell me.
4. **Silent.** Ear close to the ceiling and to the breadboard at 1000, 100 and 10 lux.
5. **No flicker.** Pi camera aimed at the ceiling, 100 µs exposure, at each level: uniform frames, no bands, no frame-to-frame wobble.
6. **Heat and stability.** One hour at 1000 lux: heatsink warm, not too hot to touch. Lux the same (within a few %) at the start and end. Repeat the three levels next day at the same numbers.

10 · Spares and the Bridgelux alternative

With the DuoStrips staying, LED spares are the spare segments you already own — a failed segment is cut off and a new one wired into the same two WAGOs, then `sweep()` re-run for that room. The electronic parts most worth a spare are already doubled in the list (second OP295, third sense resistor); the IRLZ44N is ~C$1.75 if you want one in the drawer.

The Bridgelux EB 280 mm module design (rev 2) remains fully simulated and is the fallback if the DuoStrips ever need replacing wholesale: four modules at ~C$10 each, but note that Digi-Key lists some variants only as 200-piece factory trays. Any BXEB-L0280Z 5000 K variant Digi-Key shows in stock with a minimum of 1 (E = CRI 80, G = CRI 90, S "Thrive" = CRI 98, all 19–20 V at 350 mA) is a drop-in for that circuit, which uses a 3.0 Ω sense resistor and clip-on heatsinks instead of the 2.0 Ω / bolt-on parts above.